

LECTURE 3.2



Functional Form and Dummies

ECONXXXXX Econometrics I

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PLACEHOLDER TERM

Where We Are

1. Foundations

1.1 Data and Variation

1.2 Conditional Means

1.3 Sampling Variation

2. Simple Regression

2.1 Fitting a Line

2.2 Goodness of Fit

2.3 Bias and Precision

2.4 Reporting an Estimate

3. Multiple Regression

3.1 Adding Controls

3.2 Functional Form

3.3 Joint Tests

4. Limits

4.1 Diagnosing Violations

4.2 Limits of Regression

The Second of Three Lectures on Building a Model

Lectures already given are greyed and today's is marked. Lecture 3.1 put a second variable into the equation. Today keeps the number of variables fixed and changes what each one is, and 3.3 asks which version to keep.

What Lecture 3.1 Left Unasked

Lecture 3.1 put ability into the earnings equation. The return to schooling fell from 1.6700 to 1.2538, and we could say why: ability raises schooling and ability raises pay, so the short regression attributed both to schooling.

That was an argument about which variables belong. It left the variables themselves alone. Every equation this module has written, from 2.1 onwards, says that the tenth year of schooling and the twentieth are each worth the same 1.30 euro to the same person.

Question

Which lecture decided that? And is it a claim about the Irish labour market, or a claim about a straight line?

Lecture Plan

1 Logs and Elasticities.

Take the log of a variable and its coefficient answers a different question, in proportions rather than euro.

2 Dummy Variables.

A category enters as a 0/1 column and shifts the intercept, relative to a base category you chose.

3 Interactions.

Let the slope differ across two groups, and the coefficient on schooling stops meaning what it meant.

SECTION 1

Logs and Elasticities

1. Logs and Elasticities

2. Dummy Variables

3. Interactions

One Euro Figure at Every Level of Schooling

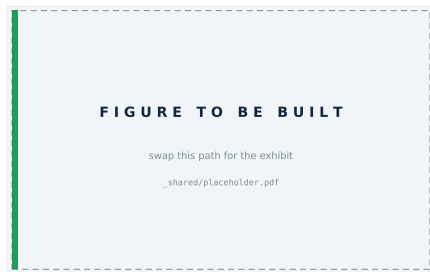


Figure 1: The same 120 people twice: earnings on schooling, then log earnings on schooling.

Two Straight Lines, Two Different Claims About Ireland

The left panel says each extra year adds the same euro figure at every level. The right panel says each extra year raises earnings by the same proportion, so its euro value rises as schooling rises. Neither is more natural than the other.

Logs Put the Return in Proportional Terms

To ask what a year of schooling is worth as a proportion of what a person already earns, put the logarithm on the left-hand side:

$$\log(w_i) = \beta_0 + \beta_1(S_i - 13) + u_i \quad (1)$$

- w_i : hourly earnings of person i , in euro
- $\log(\cdot)$: the natural logarithm
- S_i : years of schooling, centred at thirteen
- β_1 : the rise in log earnings per year of schooling
- u_i : everything other than schooling

A one-year rise in schooling adds β_1 to log earnings, and $100 \Delta \log(w) \approx \% \Delta w$, so $100\beta_1$ is a percentage (Wooldridge, 2020, eq. A.23 p.675). Fitted here, $\hat{\beta}_1 = 0.1481$ with a standard error of 0.0239, so the shortcut reads 14.81 per cent per year and the exact figure is $100(e^{0.1481} - 1) = 15.96$ per cent.

Why a Log-Log Slope Is an Elasticity

Put the logarithm on both sides and the slope is unit-free. An elasticity is defined as a ratio of proportional changes, so derive the slope from that definition rather than asserting the name:

$$\begin{aligned}\gamma_1 &= \frac{d \log(w)}{d \log(S)} = \frac{dw/w}{dS/S} && \text{because } \frac{d \log(x)}{dx} = \frac{1}{x} \\ &= \frac{dw}{dS} \cdot \frac{S}{w} = \frac{\% \Delta w}{\% \Delta S}\end{aligned}\tag{2}$$

■ γ_1 : the elasticity of earnings with respect to schooling

The first line rewrites each differential of a log as a proportional change. The second cancels the levels, and what is left has euro over euro on top and years over years underneath. Both units divide out, which is why an elasticity has none (Wooldridge, 2020, eq. A.24 p.676, and p.679). Fitted here, $\hat{\gamma}_1 = 2.0119$ with a standard error of 0.3185: one per cent more schooling goes with about 2.01 per cent more earnings, in euro, cents, years or months alike.

Four Specifications and How to Read Each One

Table 1: The same two variables under four specifications, and the sentence each slope supports.

Specification	Regression	Change in S	Say this out loud
Level–level	w on S	one year	One more year of schooling adds β_1 euro to hourly earnings.
Level–log	w on $\log S$	one per cent	One per cent more schooling adds $\beta_1/100$ euro to hourly earnings.
Log–level	$\log w$ on S	one year	One more year of schooling raises earnings by about $100\beta_1$ per cent.
Log–log	$\log w$ on $\log S$	one per cent	One per cent more schooling raises earnings by about β_1 per cent.

Notes

Only the first row is exact. The word “about” in the other three is the approximation $100 \Delta \log x \approx \% \Delta x$, made precise two frames from here.

The four rows are Wooldridge’s Table 2.3 (Wooldridge, 2020, p. 39). Our four fitted slopes are 1.5227, 19.7087, 0.1481 and 2.0119, in that order, and not one of them can be compared with the one above it. Write the log-log sentence out in full before the next

Per Cent and Percentage Point Are Not the Same Word

A percentage change is measured relative to where you started. A percentage point is the plain arithmetic difference between two percentages. When the variable is itself a rate, both are available and they are different numbers:

- A sales tax rising from 4 per cent to 6 per cent is a rise of **two percentage points**.
- The same rise is $100 \times (6 - 4)/4 = 50.0000$ **per cent**, because the base is 4.
- Michigan voters were told both figures in 1994, and both were correct (Wooldridge, 2020, p. 672).

For a variable measured in euro there is no ambiguity, because earnings are not a percentage and a percentage point change in them is undefined.

What This Costs When the Dependent Variable Is a Rate

Regress an unemployment rate on schooling in levels and the coefficient is in percentage points. Regress its logarithm and the coefficient is a percentage change of the rate itself. Unemployment going from 8 to 9 is one percentage point and a 12.5 per cent rise (Wooldridge, 2020, p. 188).

The Exact Percentage Change and When the Shortcut Fails

The shortcut $100\beta_1$ is an approximation. Difference the fitted model across a one-year rise in schooling and exponentiate, and the exact figure appears (Wooldridge, 2020, eq. 6.9 p.186):

$$\% \Delta w = 100(e^{\beta_1} - 1) \quad (3)$$

Table 2: The shortcut against the exact figure, computed at four values of the coefficient.

β_1	Shortcut $100\beta_1$	Exact	Gap
0.05	5.0000	5.1271	0.1271
0.10	10.0000	10.5171	0.5171
0.25	25.0000	28.4025	3.4025
0.50	50.0000	64.8721	14.8721

Notes

Pure arithmetic about the exponential function, to four decimal places. The gap grows with the square of β_1 , so it quadruples when β_1 doubles.

What Logs Cost: One of Our Own 120 People

The logarithm is undefined at zero and below, and the lowest earner in our standing sample is on -0.06 euro an hour. Every log specification on this deck is therefore fitted on **119 people, not 120**:

- The level-level slope, 1.5227, uses all 120 of them.
- The log-level slope, 0.1481, and the log-log slope, 2.0119, use 119.
- The person dropped is the poorest one, which is rarely the person you can spare.

Adding a euro before taking the log would keep them, but that constant sets the answer and is a choice about units rather than a repair. Years of schooling stay unlogged throughout, because a year is already a natural unit (Wooldridge, 2020, p. 187).

Our Population Was Generated as a Straight Line

The 200,000 people were built with a constant 1.30 euro return and a normal error, which is what puts one of them below zero. A log specification fitted to them is a shape we imposed, and it still fits: R^2 is 0.2476 for the log-level model against 0.3531 for the level-level one. Those two figures are not comparable, because they account for variation in two different dependent variables (Wooldridge, 2020, p. 188), and Lecture 3.3 returns to that.

A Training Coefficient of 0.08

A regression of log earnings on a training dummy returns 0.08. A colleague writes that training raises earnings by eight percentage points.

Question

Two separate errors are stacked in that sentence. Name both, and give the figure the sentence should have carried.

SECTION 2

Dummy Variables

1. Logs and Elasticities

2. Dummy Variables

3. Interactions

A Category Enters the Model as an Intercept Shift

To put something that is not a number into a regression, code it as a column of ones and zeros and let it move the intercept. Today's group is graduates, sixteen years of schooling or more: 19 of our 120 people:

$$w_i = \beta_0 + \beta_1(S_i - 13) + \delta G_i + u_i \quad (4)$$

- G_i : one if person i has sixteen or more years of schooling, zero otherwise
- δ : the vertical gap between graduates and non-graduates

Set $G_i = 0$ and the intercept is β_0 ; set $G_i = 1$ and it is $\beta_0 + \delta$. The slope on schooling is β_1 either way, so the two fitted lines are parallel and δ is the same gap at every level of schooling (Wooldridge, 2020, eq. 7.1 p.222).

Delta Is a Difference Between Two Conditional Means

$\delta = E[w \mid G = 1, S] - E[w \mid G = 0, S]$, with schooling held at the same value in both terms (Wooldridge, 2020, eq. 7.2 p.222). Nothing about least squares is different: one more column, and only its reading is

Two Parallel Lines Through One Sample

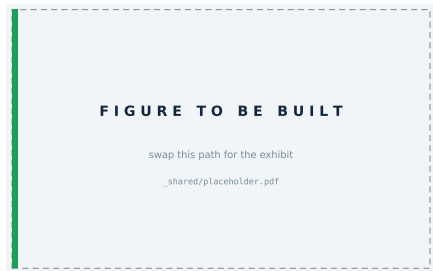


Figure 2: Graduates against non-graduates, with two parallel fitted lines and the 2.8634 gap marked.

The Vertical Gap Is the Coefficient, and It Never Widens

The two lines are parallel by construction, because the model gave both groups the same coefficient on schooling, 1.2042. If the gap in the data widens or narrows with schooling, this model cannot report that, and section 3 is the repair.

The Coefficient Is a Difference Between Two Averages

Take schooling out of that model and leave the intercept and the dummy. Least squares then has two parameters and two groups, so it reproduces each group's mean exactly. Our 101 non-graduates average 10.9550 and our 19 graduates average 19.1277, a difference of 8.1726:

$$\hat{w}_i = \underbrace{10.9550}_{\text{non-graduate mean}} + \underbrace{8.1726}_{\text{difference of the two means}} G_i \quad \text{se}(\hat{\delta}) = 1.2806 \quad (5)$$

Both fitted coefficients match the two averages to four decimal places, which is not a coincidence (Wooldridge, 2020, p. 224). So a constant and one dummy are the comparison anyone would have made without regression at all.

Schooling Cuts the Graduate Premium From 8.17 to 2.86

Put schooling back and $\hat{\delta}$ falls to 2.8634, with a standard error of 1.6594, while the slope on schooling is 1.2042. Most of the raw gap was schooling itself: graduates have more of it. That is Lecture 3.1's partialling out with a 0/1 column in place of a continuous one (Angrist & Pischke, 2015).

Every Dummy Coefficient Is a Comparison Against the Omitted Category

A graduate dummy has two categories. Highest qualification has five: Leaving Certificate, apprenticeship, certificate, degree, postgraduate. Four dummies go in and the software prints four numbers:

$$w_i = \beta_0 + \beta_1(S_i - 13) + \delta_1 Q_{1i} + \delta_2 Q_{2i} + \delta_3 Q_{3i} + \delta_4 Q_{4i} + u_i \quad (6)$$

- Q_{ji} : one if person i holds qualification j , zero otherwise
- δ_j : the gap between qualification j and the omitted one

Every one of the four is a comparison against the category you left out (Wooldridge, 2020, p. 229). Reset the base and all four printed numbers change while nothing about Ireland does. In R the base is the first level of the factor, which is alphabetical unless `relevel()` says otherwise, so spelling can decide what your table reports.

Why One Category Must Be Left Out

Put in a dummy for every one of the five qualifications as well as an intercept. Each person holds exactly one, so $\sum_{j=1}^5 Q_{ji} = 1$ for every person, which is the intercept column exactly. Now shift the coefficients by any constant c :

$$(\beta_0 + c) + \sum_{j=1}^5 (\delta_j - c) Q_{ji} = \beta_0 + \sum_{j=1}^5 \delta_j Q_{ji} - c \left(\underbrace{\sum_{j=1}^5 Q_{ji}}_{=1} - 1 \right) \quad (7)$$

The bracket is zero for every person, so every fitted value, every residual and the residual sum of squares are all unchanged. Infinitely many coefficient vectors fit identically well, so no unique least squares answer exists.

The Dummy Variable Trap

This is assumption three of Lecture 3.1, no perfect collinearity, failing exactly rather than nearly (Wooldridge, 2020, pp. MLR.3 p.80). With g categories and an intercept, $g - 1$ dummies go in. R drops one and prints the rest without comment.

Dummy Coefficients When Earnings Are Logged

With $\log(w)$ on the left, a dummy coefficient is a proportional gap and the same two readings apply, the shortcut and the exact figure (Wooldridge, 2020, eq. 7.10 p.227):

$$\text{shortcut: } 100\hat{\delta} \qquad \text{exact: } 100(e^{\hat{\delta}} - 1) \qquad (8)$$

At $\hat{\delta} = 0.08$ the shortcut gives 8.0000 and the exact figure is 8.3287, because $\log(1.08) = 0.0770$ and a coefficient of 0.0770 is what an 8.00 per cent gap actually produces. At $\hat{\delta} = -0.297$ Wooldridge's own wage equation reads 29.7 per cent as a shortcut and 25.7 per cent exactly (Wooldridge, 2020, p. 227).

Why the Shortcut Is Still Worth Printing

Reverse the base group and the exact figure is not the same number: the gap the other way round is 34.6 per cent, not 25.7. The shortcut, 29.7, sits between the two, so it is the one figure that does not require you to choose a direction first (Wooldridge, 2020, p. 228).

Three Qualification Coefficients and No Base Named

A table reports coefficients on apprenticeship, degree and postgraduate. Your co-author writes that the degree coefficient is the return to a degree.

Question

What is that number a comparison against, and what else must the table print before any sentence about it can be written?

SECTION 3

Interactions

1. Logs and Elasticities
2. Dummy Variables
3. Interactions

Interaction Terms Let the Slope Differ Across Groups

Nothing so far lets the return to schooling differ between two groups. Add the product of schooling and the dummy as one more column:

$$w_i = \beta_0 + \beta_1(S_i - 13) + \delta G_i + \gamma(S_i - 13)G_i + u_i \quad (9)$$

- $(S_i - 13)G_i$: the product of the two, one variable, one column
- γ : the difference in slope between graduates and non-graduates

The product is zero for everyone with $G_i = 0$ and equals centred schooling for everyone else, which is why it can sit beside both parent terms without repeating either (Wooldridge, 2020, eq. 7.17 p.233).

Both Parent Terms Stay In

The equal-slopes claim is $H_0 : \gamma = 0$, and it is a claim about the slope only: a gap between the groups is still allowed under it (Wooldridge, 2020, p. 234). Dropping G_i or $(S_i - 13)$ tests something nobody asked about.

Two Fitted Lines That Are Not Parallel

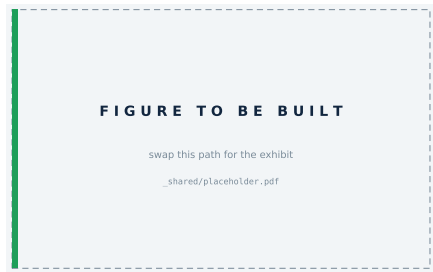


Figure 3: The same two groups, now with two fitted lines whose slopes are allowed to differ.

The Difference in Slope Is the Interaction Coefficient

The vertical gap between the lines is no longer one number. It narrows as schooling rises, and the rate at which it does so is $\gamma = -1.2068$. The graduate line is nearly flat, and the next two frames say both what that means and how little the 120 people settle it.

What the Schooling Coefficient Means Once an Interaction Is In

Substitute the two values of the dummy into the model and read the slope off each line:

$$\begin{aligned} G_i = 0 : \quad \hat{w}_i &= 11.2710 + 1.2599 (S_i - 13) \\ G_i = 1 : \quad \hat{w}_i &= (11.2710 + 7.6357) + (1.2599 - 1.2068)(S_i - 13) \end{aligned} \tag{10}$$

So the return is 1.2599 euro a year for non-graduates and $1.2599 - 1.2068 = 0.0531$ for graduates, and $\hat{\gamma} = -1.2068$ is the difference between them (Wooldridge, 2020, eq. 7.16 p.233).

Before the interaction, 1.2042 was the return for everybody. After it, 1.2599 is the return for non-graduates alone. It is not an average of the two, it is not the overall return, and the printout will not warn you that its meaning is now narrower.

This Misreading Is Not a Student Error

It is the most common error in published work using interactions. The coefficient on a main term is a partial effect evaluated at one value of the other regressor, and with a dummy that value is zero, meaning the base group (Wooldridge, 2020, p. 193).

Why Centring Schooling Makes the Intercept Interpretable

Run the same model on raw schooling instead of centred schooling, and take the difference between the two groups at a given S_i :

$$\begin{aligned}\text{raw scale: } E[w \mid G = 1, S] - E[w \mid G = 0, S] &= \delta + \gamma S_i \\ \text{centred: } E[w \mid G = 1, S] - E[w \mid G = 0, S] &= \delta + \gamma(S_i - 13)\end{aligned}\tag{11}$$

On the raw scale δ is the gap at $S_i = 0$, which is nobody: no one in the population has zero years of schooling. On the centred scale $\hat{\delta} = 7.6357$ is the gap at thirteen years, the Leaving Certificate, and $\hat{\beta}_0 = 11.2710$ is expected hourly earnings there for a non-graduate.

Centring Relocates Two Coefficients and Nothing Else

Every fitted value, every residual and γ itself are identical under both (Wooldridge, 2020, p. 193). Centring is not a statistical repair. It reports two of the coefficients at a point you can defend, and their standard errors at that point come with them.

The Standard Error of the Slope for the Other Group

Reading 1.2599 and 0.0531 off a table and concluding that the two groups differ is the error this lecture exists to prevent. The claim is about γ , so it has to be tested on γ .

The return for graduates is $\hat{\beta}_1 + \hat{\gamma}$, and the spread of a sum is not the sum of the spreads. Lecture 1.2's algebra of variances gives:

$$\text{Var}(\hat{\beta}_1 + \hat{\gamma}) = \text{Var}(\hat{\beta}_1) + \text{Var}(\hat{\gamma}) + 2 \text{Cov}(\hat{\beta}_1, \hat{\gamma}) \quad (12)$$

- $\text{Cov}(\hat{\beta}_1, \hat{\gamma})$: how the two estimates miss together across samples

That covariance is not on the printout, so the standard error cannot be assembled from the two printed ones. Refit with graduates as the base and read $\hat{\beta}_1 + \hat{\gamma}$ straight off, with its standard error attached (Wooldridge, 2020, pp. 194, p.229).

1.2599 Against 0.0531, and t Equal to Minus 0.96

The two fitted slopes look nothing alike, and the difference between them is not distinguishable from zero: $\hat{\gamma} = -1.2068$ has a standard error of 1.2566, so $t = -0.96$ and Lecture 2.4's rule does not reject equal slopes. An interaction is estimated off a difference between two subgroup slopes, one of them fitted on 18 people, so it is far less precise than either

The Return to Schooling for Women

Log earnings on centred schooling, a dummy for women, and their interaction. You are asked for the return to schooling for women.

Question

Which of the three coefficients answers it, and how would you obtain the standard error that belongs beside your answer?

What Would Have To Be True?

This time the question is asked of the shape, not of a number. The straight line, the logarithm and the graduate cut at sixteen years were all chosen by us before anything was fitted, and the graduate dummy is built out of schooling itself, so the two groups barely overlap in it.

Question

What would have to be true for this number to mean what we just said it means? And what is the plausible story where it isn't?

Application

Fit all four specifications to the same people and write each slope out as a sentence with units before reading the next one. Level-level gives 1.5227 with $R^2 = 0.3531$; level-log 19.7087 with 0.3309; log-level 0.1481 with 0.2476; log-log 2.0119 with 0.2543. The last two are fitted on 119 people.

Question

Answer before anything is run. Which of the four will report the highest R^2 , and does the winner tell you which shape generated the data?

The lab adds `log()`, `factor()` and `relevel()`, and `y ~ x * d` as shorthand for a model carrying both parent terms and their product.

Four Fitted Shapes Over One Scatter

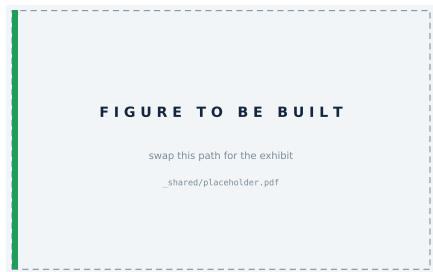


Figure 4: Predicted earnings under all four specifications, drawn back onto the euro scale.

They Agree Where the Data Are and Separate Where They Are Not

All four are drawn on the level scale so that they can be compared at all. The shaded band holds the middle eighty per cent of the sample's schooling. Inside it the four curves are close; outside it the specification decides the answer.

Summary: Logs, Dummies and Interactions

1 Logs and Elasticities.

A log puts the coefficient in proportions: 0.1481 per year, 2.0119 per cent per cent, and one of our 120 people dropped.

2 Dummy Variables.

A dummy shifts the intercept by a difference in two means, 8.1726 raw and 2.8634 once schooling is held fixed.

3 Interactions.

The product term lets the slope differ, 1.2599 against 0.0531, and $t = -0.96$ says our 120 people do not settle that.

Next Lecture

Today produced a block of four qualification dummies and an interaction with $t = -0.96$, and no tool for deciding whether either belongs. Lecture 3.3 supplies it:

- **Testing Restrictions:** asking about four coefficients at once, which four separate t-statistics cannot do.
- **Choosing a Specification:** what adjusted R^2 charges for a column, and why 0.3531 and 0.2476 are not a comparison.
- **Multicollinearity:** what correlated columns do to a standard error, which is where today's 1.2566 comes from.

The first question in 3.3 is which coefficient gives the gap between two qualifications that are both in the model.

Before Lecture 3.3

1 Read:

Wooldridge (2020), section 2-4 and Table 2.3 on the four forms, then 7-2 and 7-4 on dummies and interactions.

2 Apply:

Problem set 3, covering 3.1 and 3.2. It asks for one results table written out in prose with correct units in every sentence.

3 Play:

The app. Switch the specification and watch the fitted shape and the printed sentence change together.

Thank you

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BACKUP

Appendix

Derivation: Where the Exact Percentage Change Comes From

The body frame stated $\% \Delta \hat{w} = 100(e^{\beta_1} - 1)$. It is the fitted model differenced across one year and exponentiated:

$$\log \hat{w}(S+1) - \log \hat{w}(S) = \beta_1 \implies \frac{\hat{w}(S+1)}{\hat{w}(S)} = e^{\beta_1} \implies \frac{\Delta \hat{w}}{\hat{w}} = e^{\beta_1} - 1 \quad (13)$$

The Shortcut Is the First Term of a Series

$e^{\beta_1} - 1 = \beta_1 + \beta_1^2/2 + \dots$, so the shortcut drops everything past the first term and the leading error is $\beta_1^2/2$. That is 0.00125 at $\beta_1 = 0.05$ and 0.125 at $\beta_1 = 0.50$, which is the shape of the gap column on the body frame (Wooldridge, 2020, p. 186).

Derivation: Why the Centring Constant Cancels From the Slopes

Replacing S_i by $S_i - 13$ throughout rewrites the intercept and the dummy coefficient and leaves both slopes alone:

$$\beta_0^{\text{raw}} + \beta_1 S_i + \delta^{\text{raw}} G_i + \gamma S_i G_i = \underbrace{(\beta_0^{\text{raw}} + 13\beta_1)}_{\beta_0} + \beta_1 (S_i - 13) + \underbrace{(\delta^{\text{raw}} + 13\gamma)}_{\delta} G_i + \gamma (S_i - 13) G_i \quad (14)$$

Two Coefficients Are Relabelled, Two Are Not

β_1 and γ appear unchanged on both sides, so no slope, fitted value or residual differs. Only the intercept and the dummy coefficient are relabelled, which is the whole of what centring does (Wooldridge, 2020, p. 193).

Notes and Tools

- **The App:** the specification app, with the four forms and the interaction switchable.
- **The Lecture in One Line:** $\text{lm}(\log(w) \sim I(S - 13) * D)$, which carries both parent terms.
- **This Lecture's Lab:** `log()`, `factor()`, `relevel()`, and $I(S - 13)$ for centring without touching the data.

References

Angrist, J. D., & Pischke, J.-S. (2015). *Mastering 'metrics: The path from cause to effect*. Princeton University Press.

Wooldridge, J. M. (2020). *Introductory econometrics: A modern approach* (7th ed.). Cengage Learning.